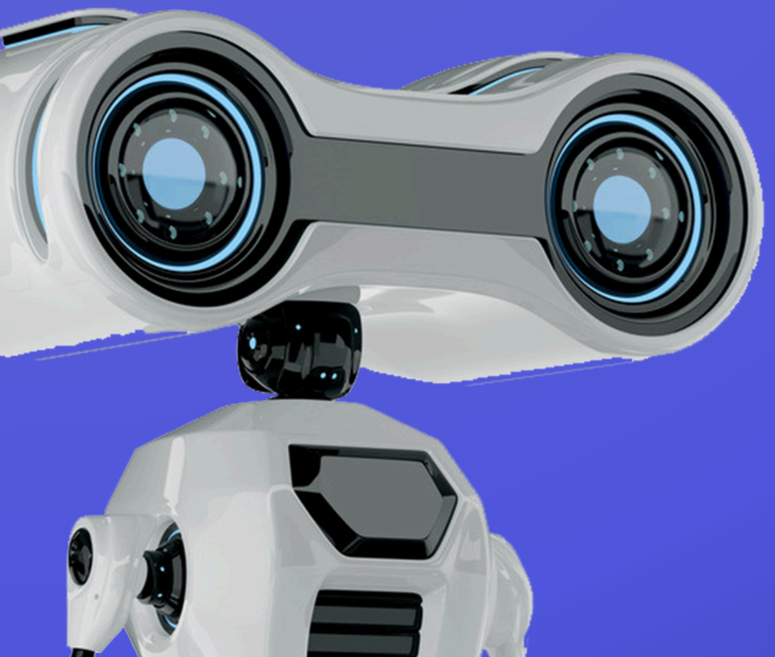
Machine Learning-Based Message Prioritization for Congestion-Aware MAVLink UAV Communication

An intelligent framework for adaptive message prioritization and congestion-aware transmission in UAV communication networks.

Start Presentation



Introduction

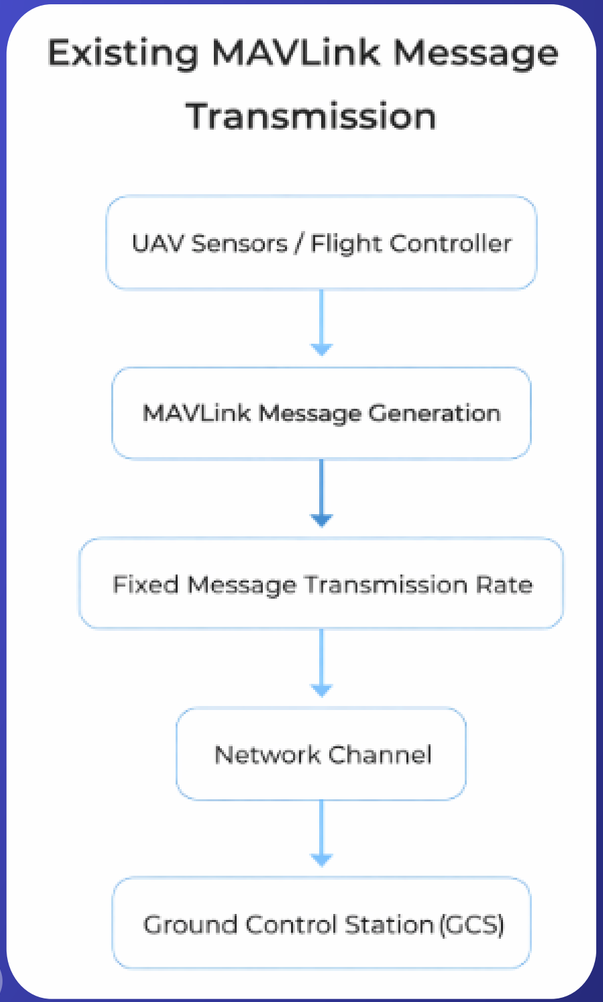
Unmanned Aerial Vehicles (UAVs) rely on continuous communication between drones and Ground Control Stations.
Communication includes:

- telemetry data
- navigation information
- control commands
- emergency alerts

These messages are transmitted using the MAVLink.
It is responsible for transmitting:

- telemetry data
- navigation information
- command messages
- system status

However, when many messages are transmitted simultaneously, network congestion can occur, affecting communication reliability.
MAVLink is a lightweight communication protocol used in UAV systems.



Problem Statement

Problem in UAV Communication

In UAV networks using MAVLink:

- Multiple telemetry, control, and sensor messages are transmitted simultaneously
- Limited network bandwidth leads to communication congestion
- All messages compete for transmission

This may cause:

- Delay in critical UAV commands
- Packet loss of important data
- Reduced reliability during flight operations

Research Objectives

The objective of this research is to:

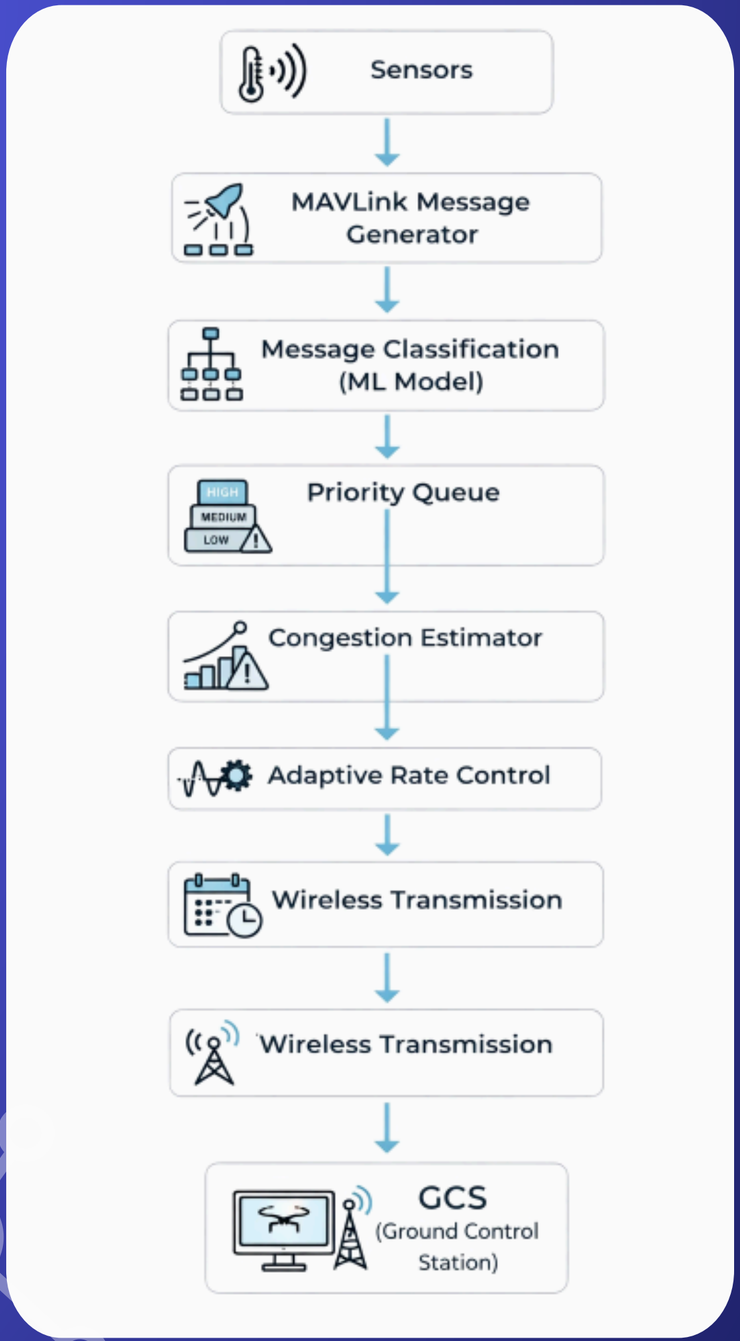
1. **Develop an intelligent system for MAVLink message prioritization**
2. **Use machine learning to classify message importance**
3. **Implement congestion-aware message scheduling**
4. **Ensure high priority UAV messages are transmitted first**

THE PROPOSED SYSTEM INTRODUCES AI-DRIVEN MESSAGE PRIORITIZATION IN UAV COMMUNICATION.

Dataset Description

The dataset represents MAVLink message scenarios with features including:

Feature	Description
message_type	type of UAV message
source	message sender
payload_size_bytes	size of message
send_frequency_hz	transmission rate
network_congestion	network load
battery_level_percent	UAV battery status
mission_phase	flight stage
emergency_flag	emergency condition



ML Model for Message Prioritization

Model: Random Forest classifier used to classify MAVLink message priority.
Why Random Forest

- Handles multiple features
- Reduces overfitting (ensemble learning)
- Robust classification

Priority Levels
High: Emergency messages, low battery alerts, landing commands
Medium: GPS updates, navigation telemetry
Low: Camera frames, non-critical data packets

ML Evaluation

Congestion-Aware Scheduling

- High: transmit immediately
- Medium: normal transmission
- Low: reduced during congestion

➡ **Ensures critical UAV messages are prioritized**

Model Evaluation

- Accuracy
- Precision
- Recall
- F1-Score

Example Prediction

- Type: ALERT | Battery: 15% | Phase: LANDING
- Congestion: 0.8 | Emergency: 1

Predicted Priority: HIGH

Congestion and Message rate control in MAVLink

What is congestion?

- In UAV Communication systems, multiple messages are transmitted simultaneously over bandwidth-limited wireless links.
- When the messages inflow exceeds network capacity, then congestion occurs

Effects of Congestion

- Packet loss
- Increased communication delay
- Unreliable command delivery
- Degradation of Quality of Service (QoS)

Lets see how network congestion is estimated:

It is estimated using different network level indicators,

- Packet Loss Ratio
- Queue Utilization
- Bandwidth Utilization

Generally Congestion (c) is calculated on the scale of 0 to 1

where $0 <= c <= 1$

$c = 0$ means no congestion

and $c = 1$ means high congestion

If congestion occurs, then we can use the proposed method to decrease the congestion and avoid the packet loss.

Priority-Aware Adaptive Message Scheduling

Our aim is to control the congestion in the network and adjust the transmission rate of the packets.

When congestion occurs, message transmission rate is adjusted dynamically, where

- high priority messages have minimal reduction rate
- medium priority messages have moderate reduction rate
- low priority messages have significantly reduced transmission rate to ensure the reliable delivery of the messages

This transmission reduction rate is done based on a heuristic formulae.

Proposed Heuristic Rate Control Formula

$$R_{new} = R_{base} \times (1-c(1-W))$$

Where R_{new} is new transmission rate
R_{base} is old transmission rate
c is congestion of the network
W is the weight of the message basket

where priority weights are
$W_{High} = 0.9$
$W_{medium} = 0.6$
$W_{low} = 0.3$

After calculating the new transmision rate, we can update transmission rate using the command "MAV_CMD_SET_MESSAGE_INTERVAL"